

SARS-CoV-2 Immuno-Vaccinology

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Systematic Review: CoV-2

Contents:

- Overview CoV-2
- Literature: **in-silico** 2020 – 2025
- Immuno-Informatics:
 - Epitopes, Vaccines, Immuno-Simulations
- Pharmacology
 - Docking: Drugs, Phytocompounds

Overview: Human CoVs

Upper Respiratory Tract (URT):

- **Alpha-CoVs:** 229E, NL63;
- **Beta-CoVs:** OC43, KHU1;
- 15% of **common colds**: autumn/winter season;
- Usually **NO** pneumonia;

Lower Respiratory Tract (LRT):

- **Beta-CoVs:** SARS-CoV, SARS-CoV-2, MERS;

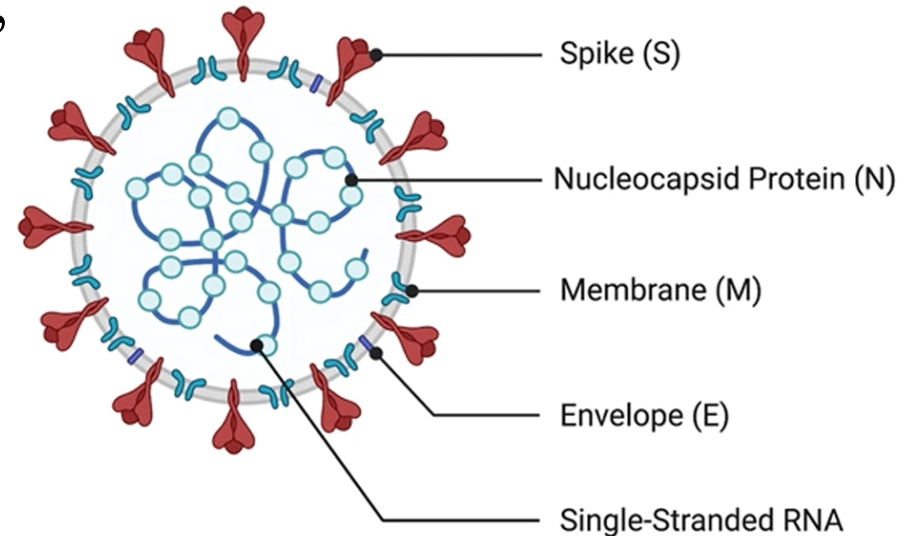
Overview: Human CoVs

Structure:

- EM: crown-like shape (corona);
- Enveloped;

Surface proteins:

- **Membrane protein (M);**
- **Spike:** > 1000 AA;
- **Envelope (E):** < 100 AA;



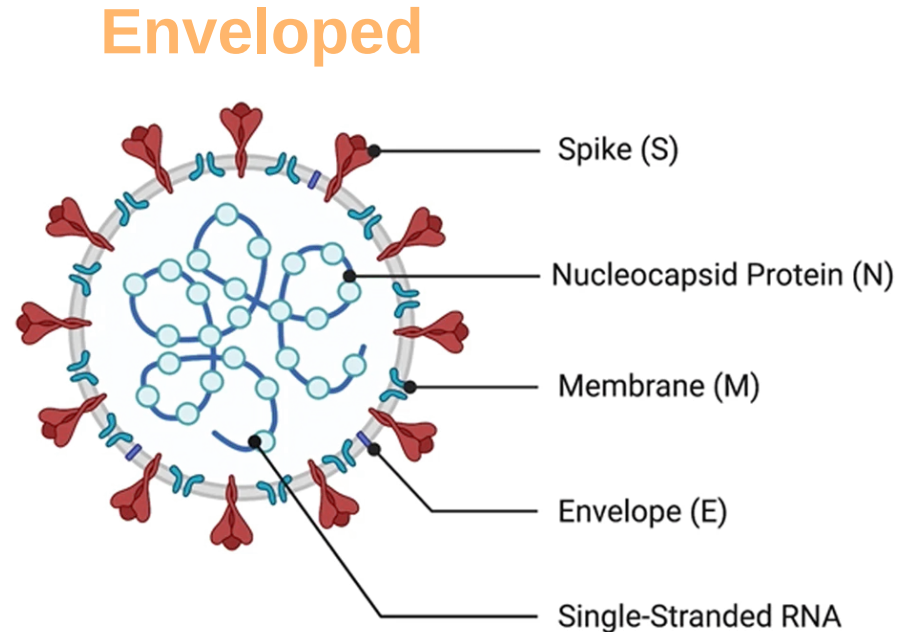
Overview: Human CoVs

Neutralising Antibodies:

- Only against: **S**, **M**, **E**; (*)
- **Cannot bind:** **N**, **NSPs**;

Surface proteins:

- **Membrane protein (M)**;
- **Spike:** > 1000 AA;
- **Envelope (E):** < 100 AA;



* E may be too small (< 100 AA); also part of it is trans-membrane;

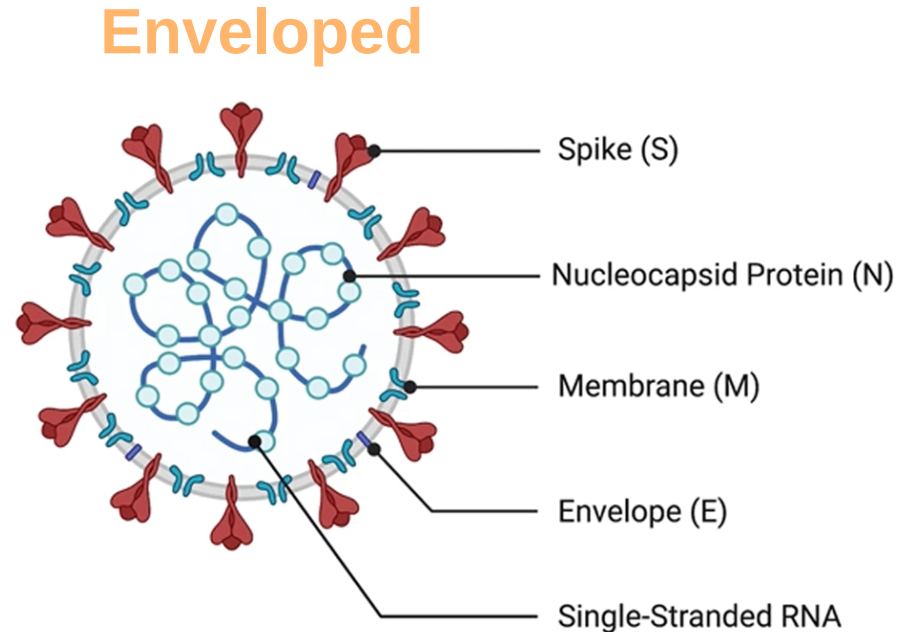
Overview: Human CoVs

Neutralising Antibodies:

- Only against: **S**, **M**, **E**;
- **Cannot bind:** **N**, **NSPs**;

Anti-N Abs:

- Cell lysis => free N protein =>
=> Ig against **linear** N epitopes;
- **NO neutralisation!**
- May still be useful: Diagnostic assays; (#)



But NOT as vaccine that induces antibodies;

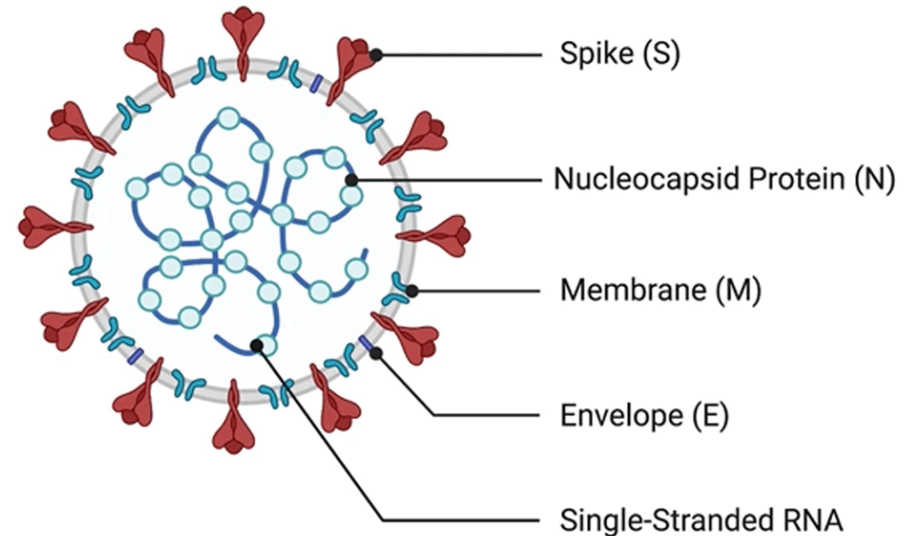
Overview: Human CoVs

Genome:

- Single stranded **RNA**;
- Plus-sense **RNA**;
- 30 kb (single molecule);

Genes:

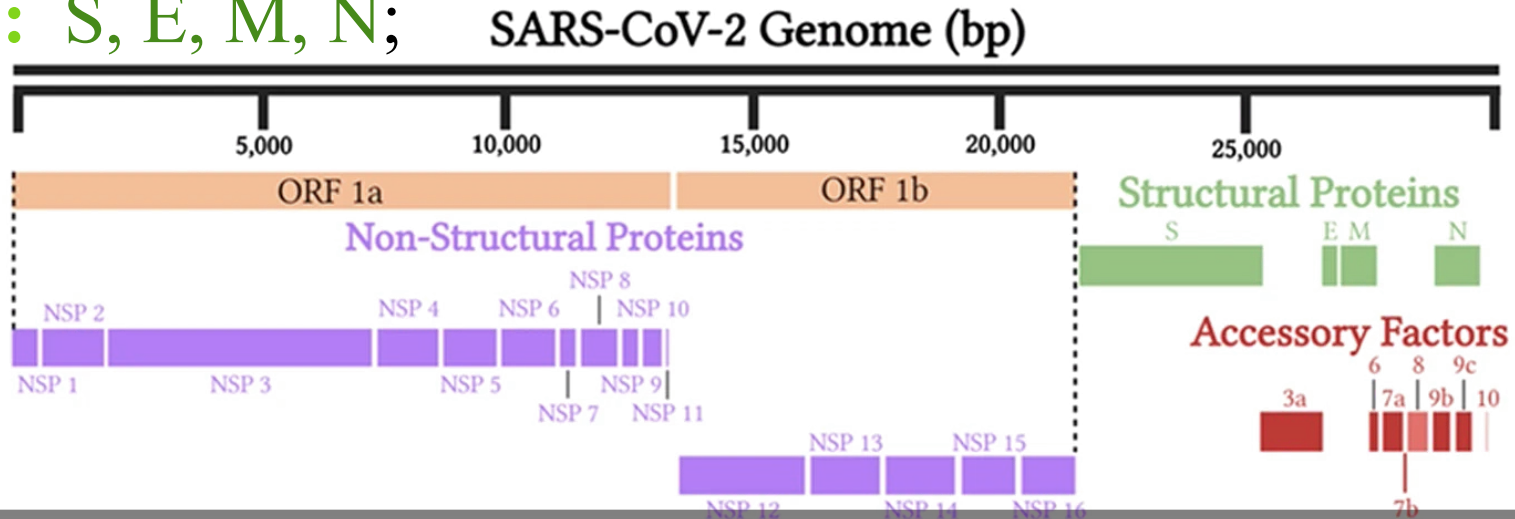
- NSPs & Structural proteins;



Overview: SARS CoV-2

Genome:

- **NSP:** ORF1a/b **Polyproteins**, others (Accessory);
- **SP:** S, E, M, N;



SP = Structural Proteins;

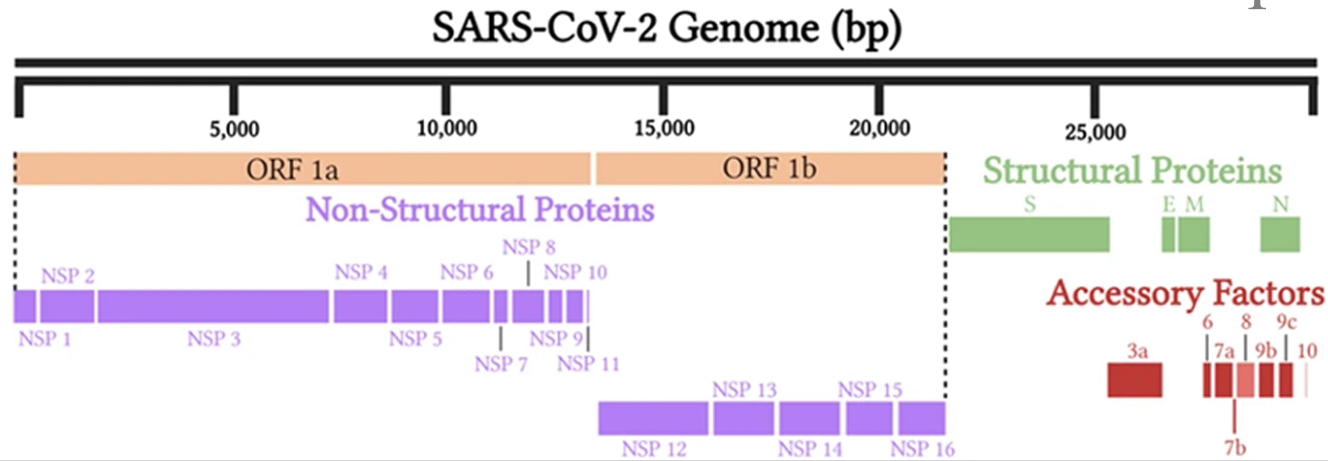
Overview: SARS CoV-2

Polyproteins: ORF1a/b

- Cleaved by **proteases**:
into 10 & 5 **proteins**;

Proteases:

- Main Protease: **Mpro**;
- Papain-Like Pro: **PLpro**;



Mpro = CLpro
Chymotrypsin-like

Literature Review: CoV-2

PubMed Query:

- SARS CoV-2;
- Immuno-informatics or in-silico;

Refinement:

- Manually => Pharmacology, Vaccinology, Other;
- NO loss of any articles;

Literature Review: CoV-2 / in-silico

PubMed Articles:

	2020	2021	2022	2023 2025
Total	600	1,128	1,027	1,242
Vaccine	90	107	89	312
Pharma	339 *	450 *	276 *	ND
ND*	70 *	392 *	612 *	ND

Difference: Other categories;

* Mostly Pharma (NO vaccine); ** ND: Not yet done;

Literature Review: CoV-2 / in-silico

PubMed Articles:

	2020	2021
Total Vaccinology	90	107
Epitopes	31	34
Vaccines #	50	64
Other	9 *	9 *

Literature Review: CoV-2 / in-silico

PubMed Articles:

Journals [#]	2020	2021
J Biomol Struct Dyn.	8	33
Sci Rep, Front Immunol, Comput Biol Med, Molecules	5 + 5 (2 + 1)	8 + 5 8 + 7
Vaccines (Bas.), J Med Virol, Inform Med Unlock, F1000Res	4 x 4	(more Ph.) 5, 3, 3, 1
Preprints ^{\$}	5+	5+

Literature Review: CoV-2 / in-silico (only 2020)

Target Proteins:

- S: 24 (only S) + 15 (in combination);
- M: 16 (mostly in combination);
- Others: N (12), E, NSPs;

Epitope types:

- HLA-I, HLA-II;
- LB: linear & discontinuous;

Literature Review: CoV-2 / in-silico (only 2020)

CoV-2 Genomes:

- **Single strain:** possibly most; (*)
- Consensus sequence from **multiple strains**;
- Example PMID 32832263:
 - Consensus: 475 strains;
 - Epitopes: 13 HLA-I & 10 HLA-II;
 - Source: S, M and other proteins;

Literature Review: CoV-2 / in-silico

Vaccine Construct:

- CTL Epitopes;
- HTL Epitopes;
- B Epitopes;
- Adjuvants: beta-Defensin, PADRE, other sequences;
- Linkers: AAY, GPGPG, EAAAK, ...;

Example:

- PMID: 32962156
- bDef – EAAAK – (CTL/AAY)₃₅ – (HTL/GPGPG)₂₁ – bDef
- Total: 1057 aa;

Literature Review: CoV-2 / in-silico

Docking:

- TLR: 3 & 4 most frequent (at least 23 & 14);
- Other: TLR 2,5,7,8,9 (a few);
- HLA molecules;
- Constructs: ~ 100 aa – 1057 aa; (some heroic dockings)
- Compare Hb: each chain 141 – 146 aa;

Literature Review: CoV-2 / in-silico

Docking:

- TLRs & HLA;
- Constructs: ~ 100 aa – 1057 aa; (some **heroic dockings**)
- Compare Hb: each chain 141 – 146 aa;

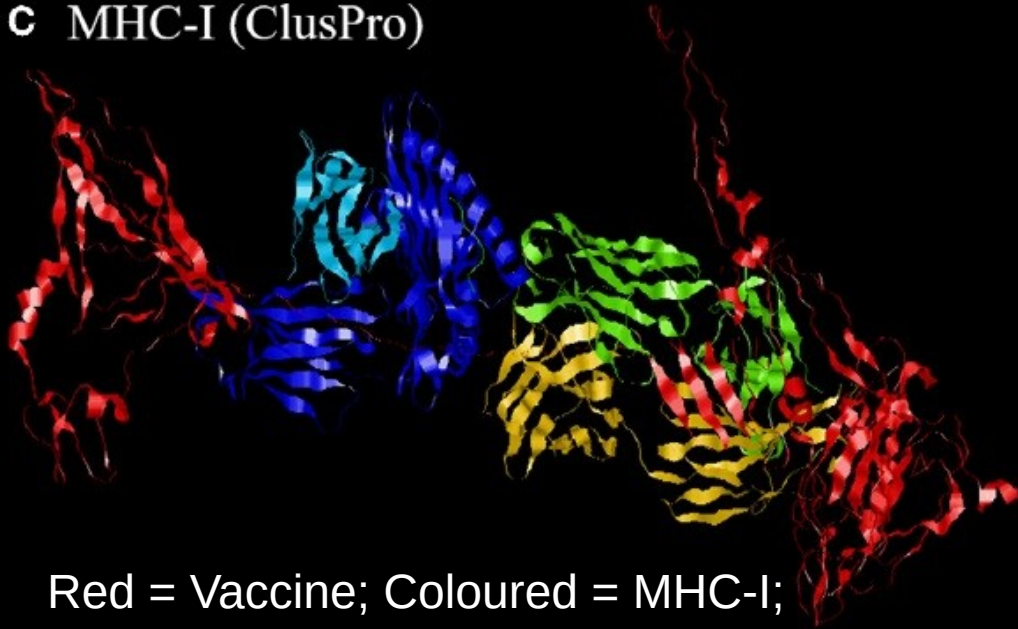
Recommendations:

- **India & China** should probably prioritize development of infrastructures for *protein NMR*;
- Build first a solid foundation for protein structures;

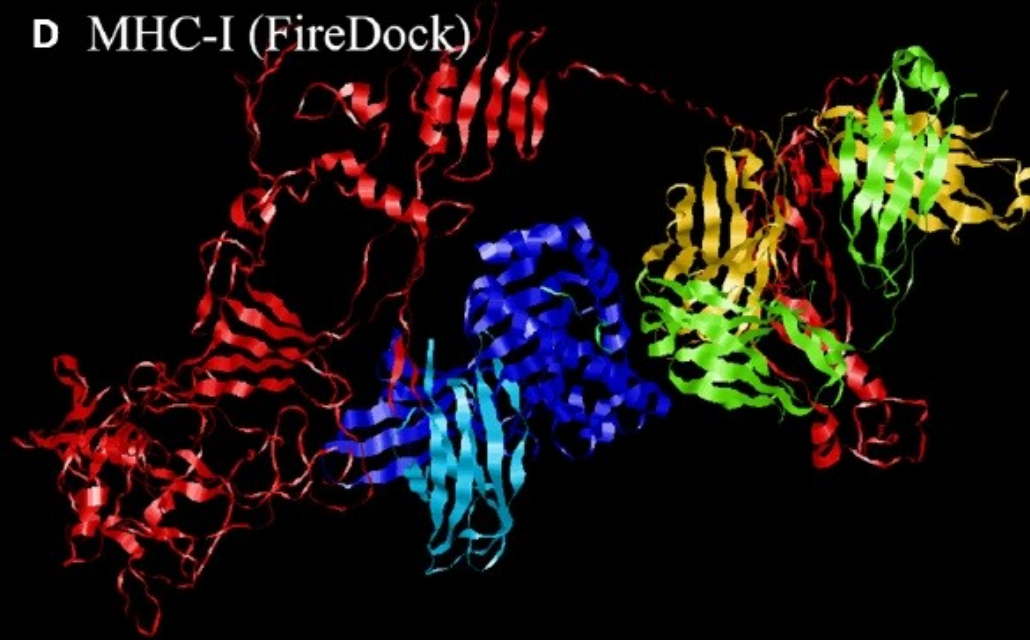
Literature Review: CoV-2 / in-silico

Docking: MHC-I + whole protein Vaccine

C MHC-I (ClusPro)



D MHC-I (FireDock)

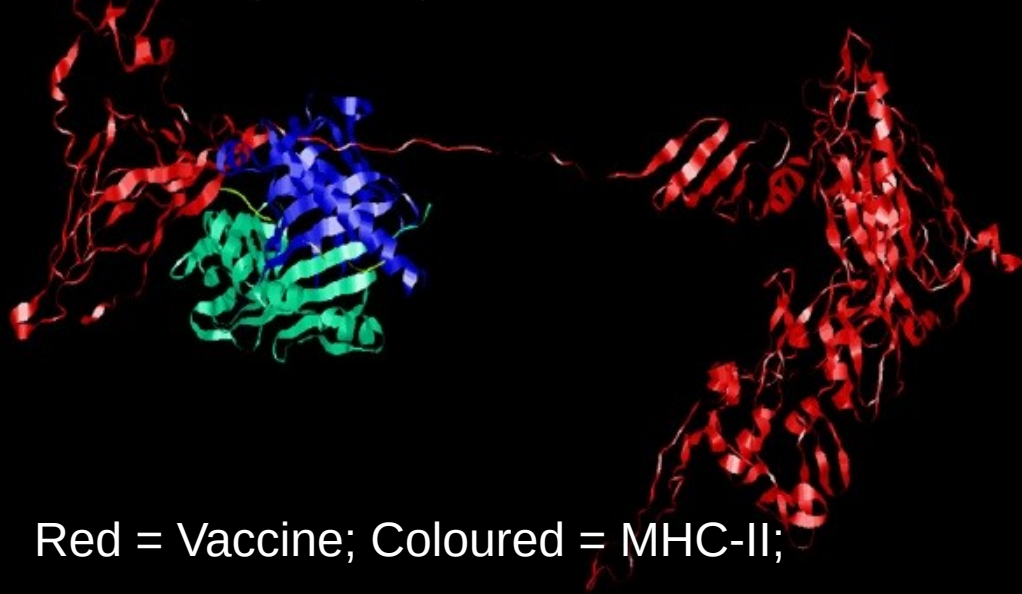


Red = Vaccine; Coloured = MHC-I;

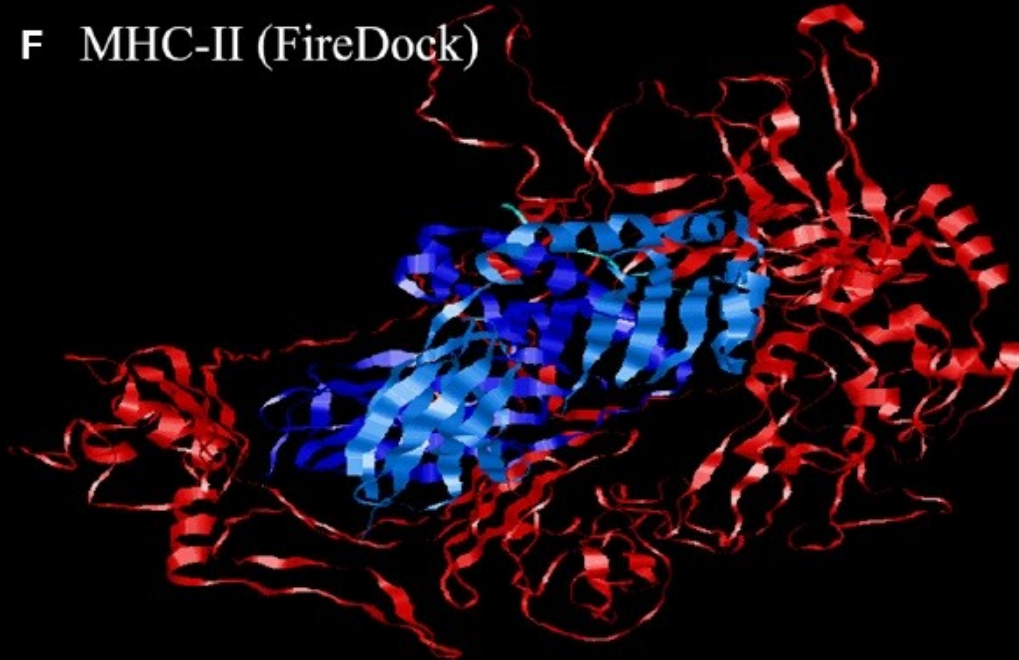
Literature Review: CoV-2 / in-silico

Docking: MHC-II + whole protein Vaccine

E MHC-II(ClusPro)



F MHC-II (FireDock)

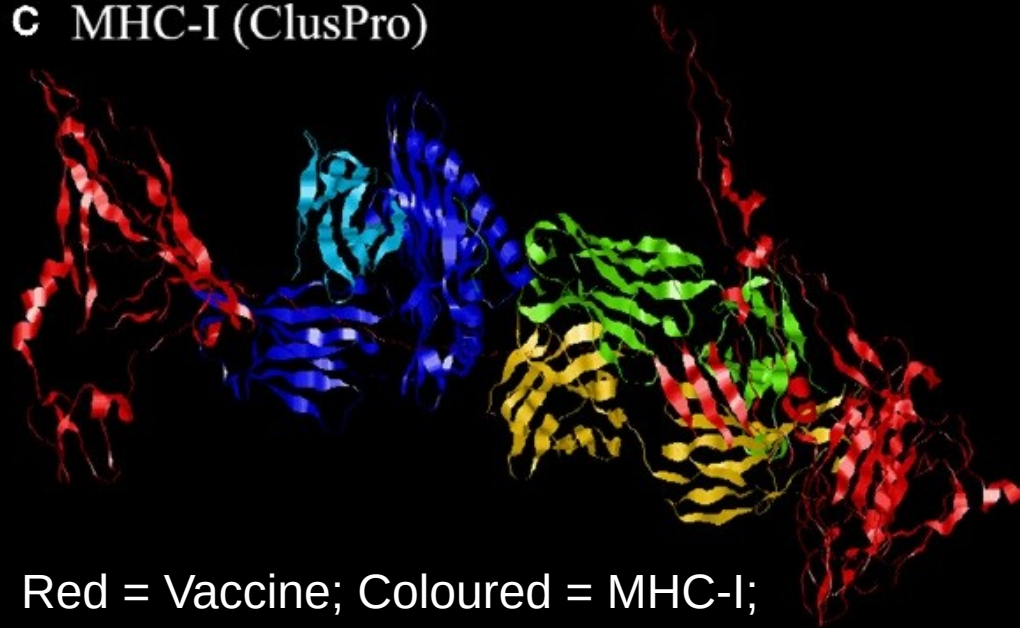


Red = Vaccine; Coloured = MHC-II;

Literature Review: CoV-2 / in-silico

Docking: MHC-I + whole protein Vaccine

c MHC-I (ClusPro)



Red = Vaccine; Coloured = MHC-I;

Is it possible to dock almost any protein?

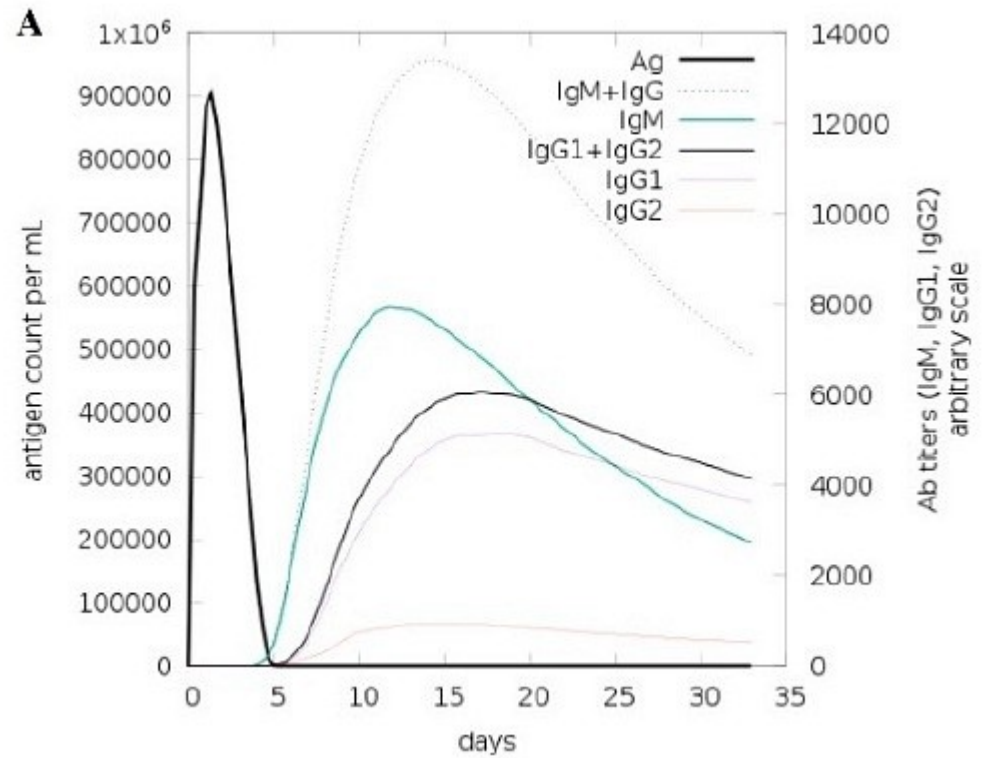
- Interesting docking exercise;
- Immunologically: **absolute bollocks**;
- **Frontiers in Immunology!**

PMID: 32849643; Front. Immunol. 2020;
For another example see: PMID 32616763; Sci Rep. 2020;

Literature Review: CoV-2 / in-silico

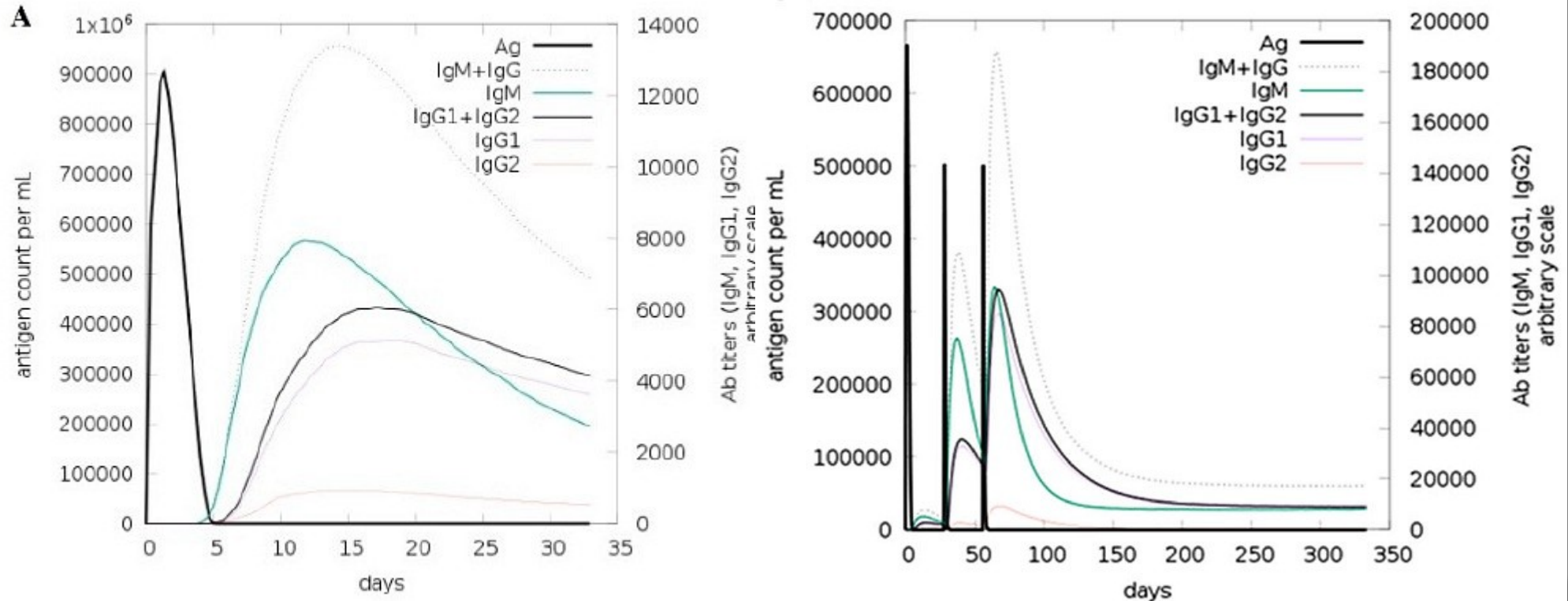
Immune Simulations:

- C-ImmSim 10.1 (2020)
- Ig sub-classes
- Cells: Tc, Th, LB
- Cytokines



Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies



* Black = Antigen/Vaccine;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!

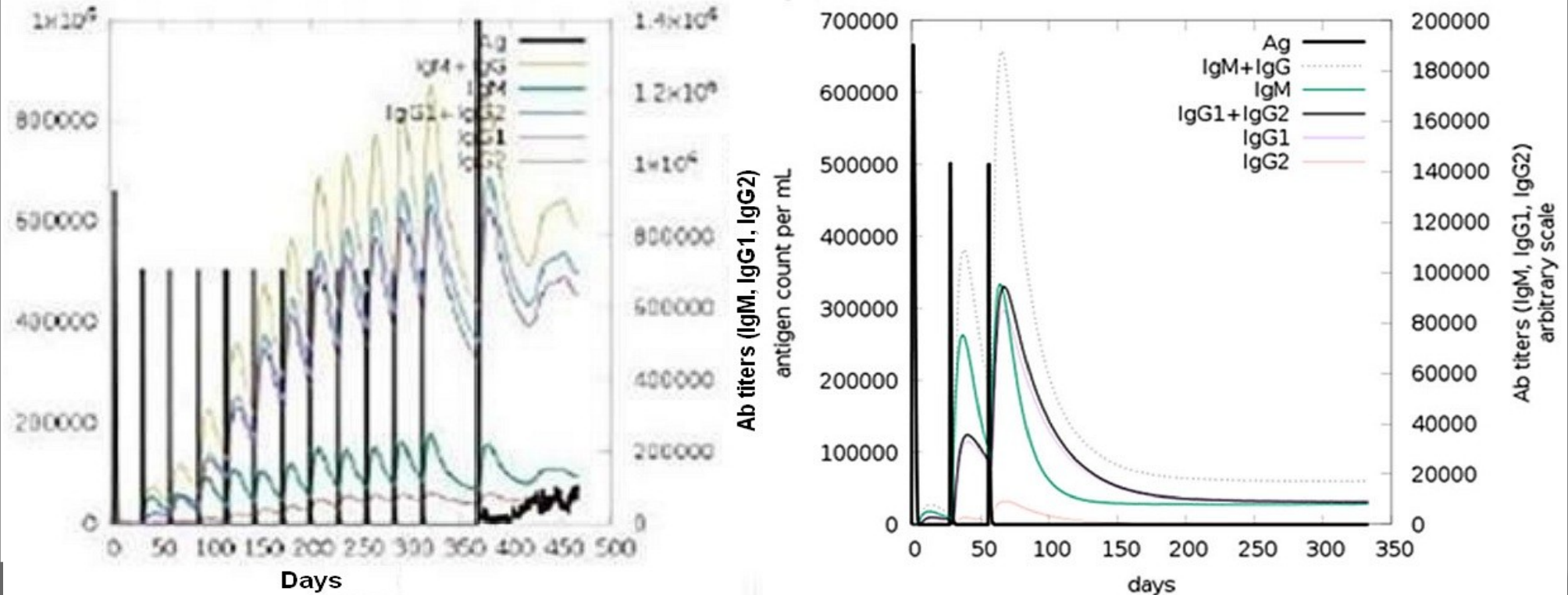
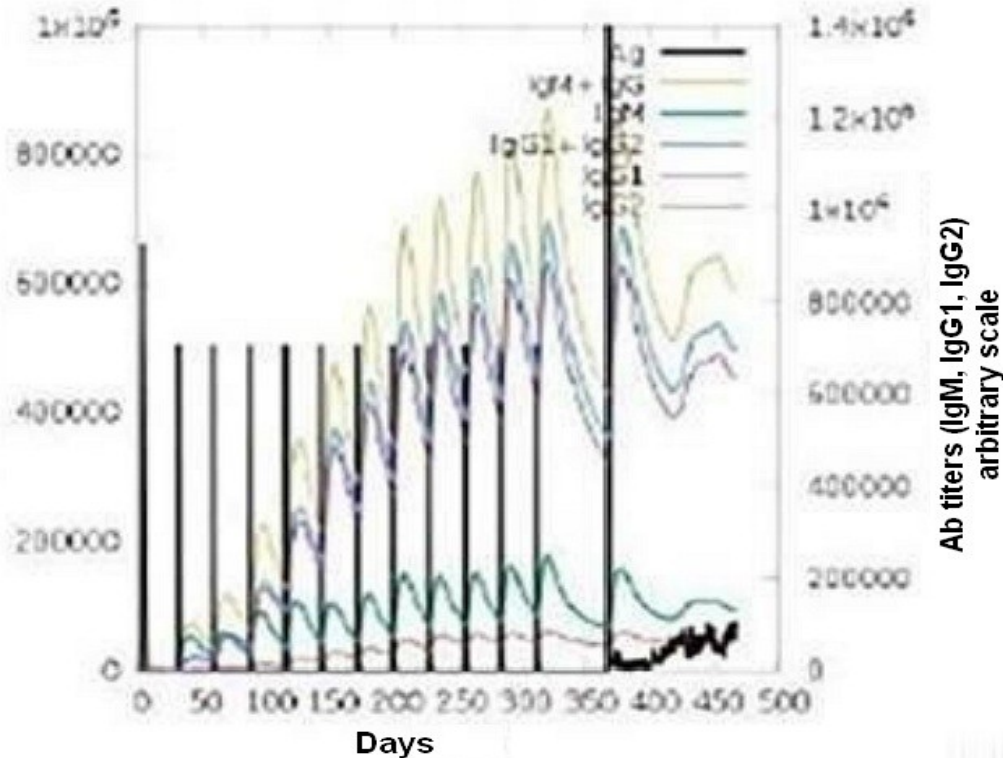


Fig 1 - Heroic Vaccination Strategy!

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



Only Pre-print:

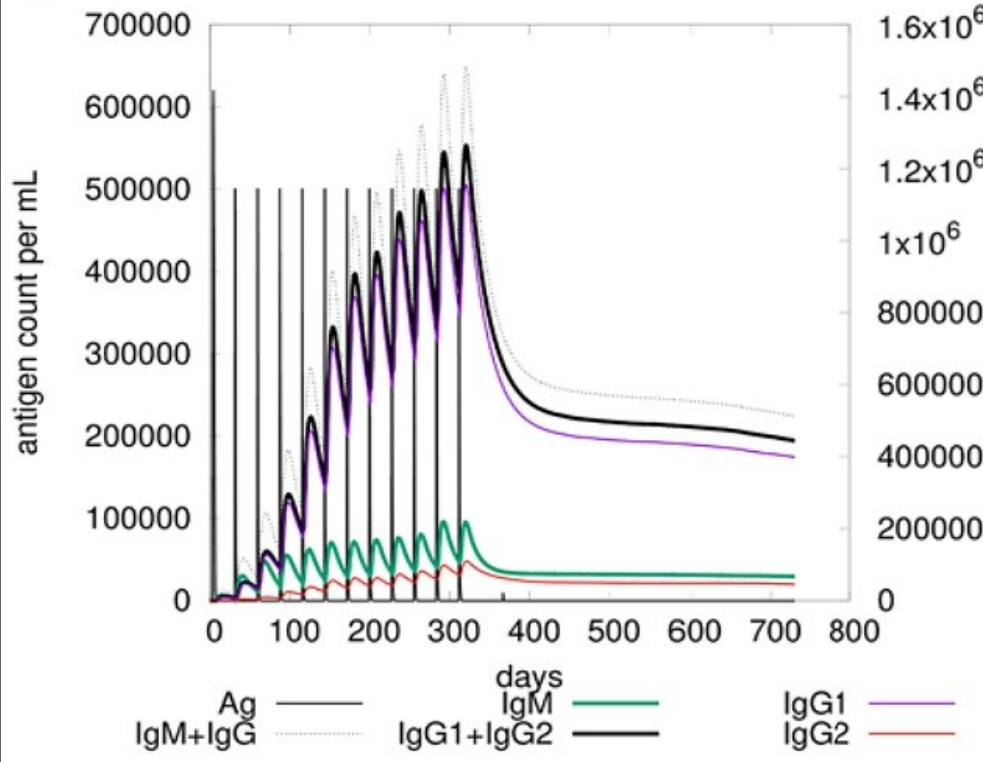
- 12 x Vaccine;
- Viral challenge afterwards;

Immuno-informatics approach for multi-epitope vaccine designing against SARS-CoV-2. bioRxiv 18-Aug-2020.

<https://doi.org/10.1101/2020.07.23.218529>;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



BUT this is published:

- 12 x Vaccine;
- *Viral challenge* afterwards;

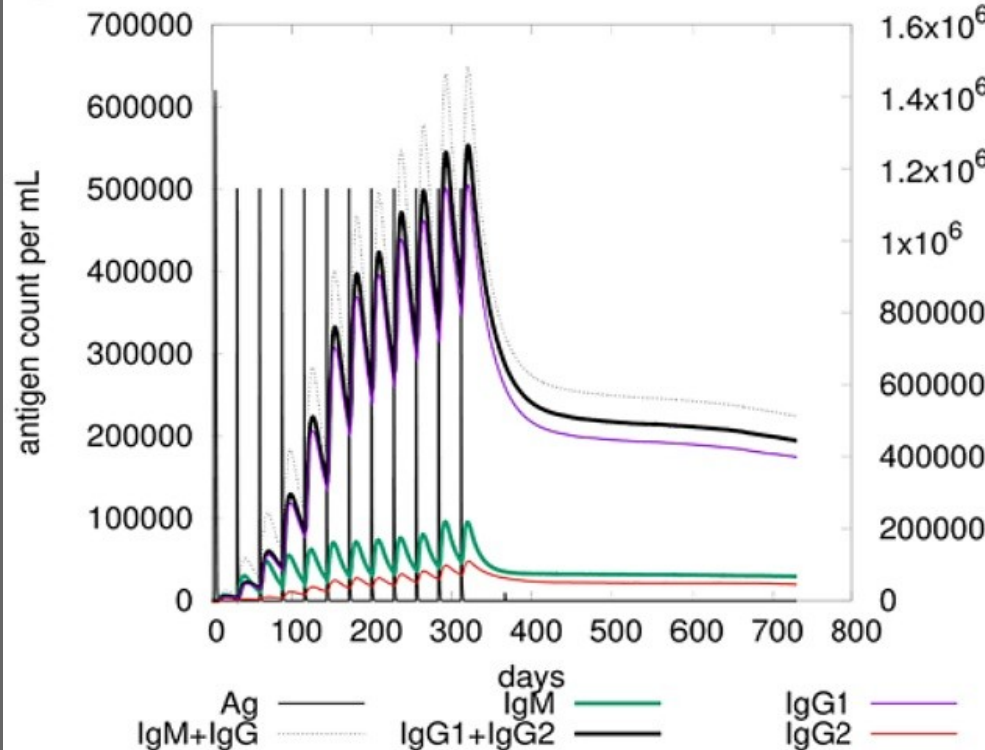
A candidate multi-epitope vaccine against SARS-CoV-2. Sci Rep. 2020 Jul 2; 10.

<https://doi.org/10.1038/s41598-020-67749-1>;

Viral Challenge: probably small tick at 365 days;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



BUT this is published:

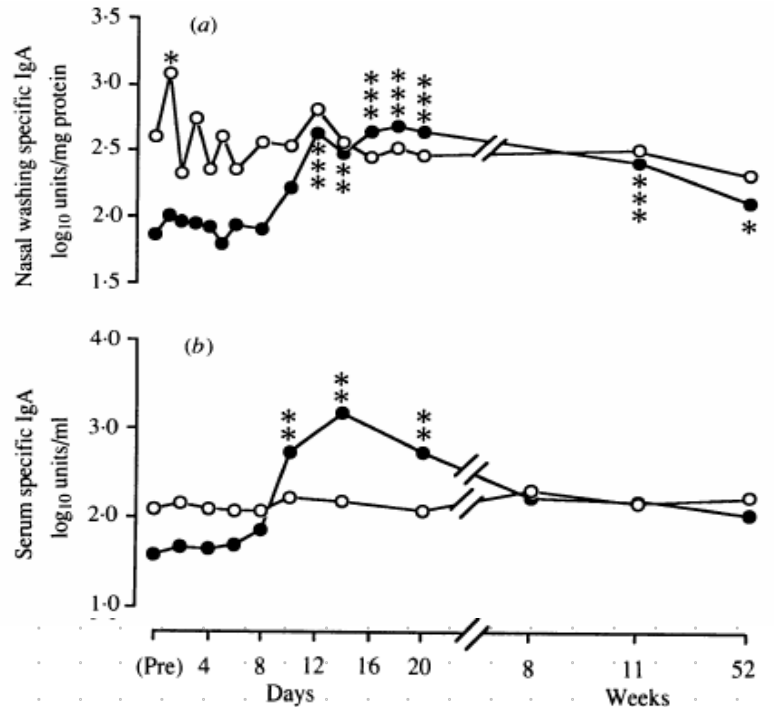
- 12 x Vaccine;

Recommendation:

Indian Gov can rehearse strategy to vaccinate each month the entire population for 12 consecutive months;

Literature Review: CoV-2 / in-silico

Real IgA: Mucosa vs Serum



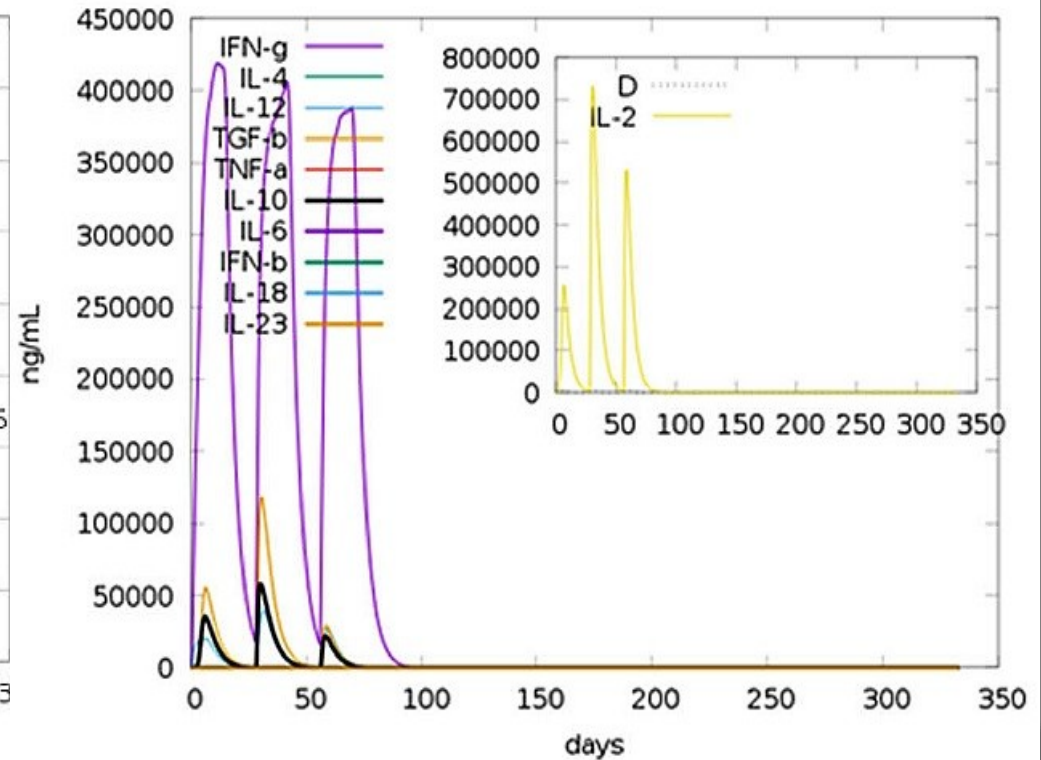
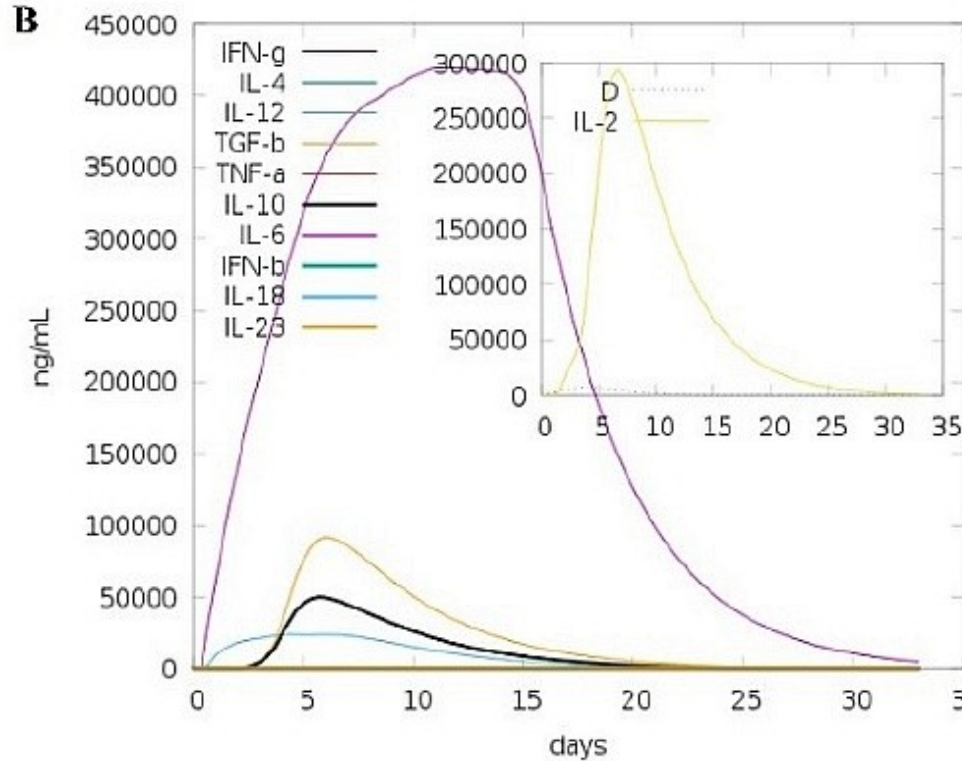
Volunteers inoculated with CoV 229E:

- Infected: 10;
- Uninfected: 5
- Pre-existing IgA;
- Surge in Mucosal IgA;

In-silico Literature:
NO simulations of IgA!

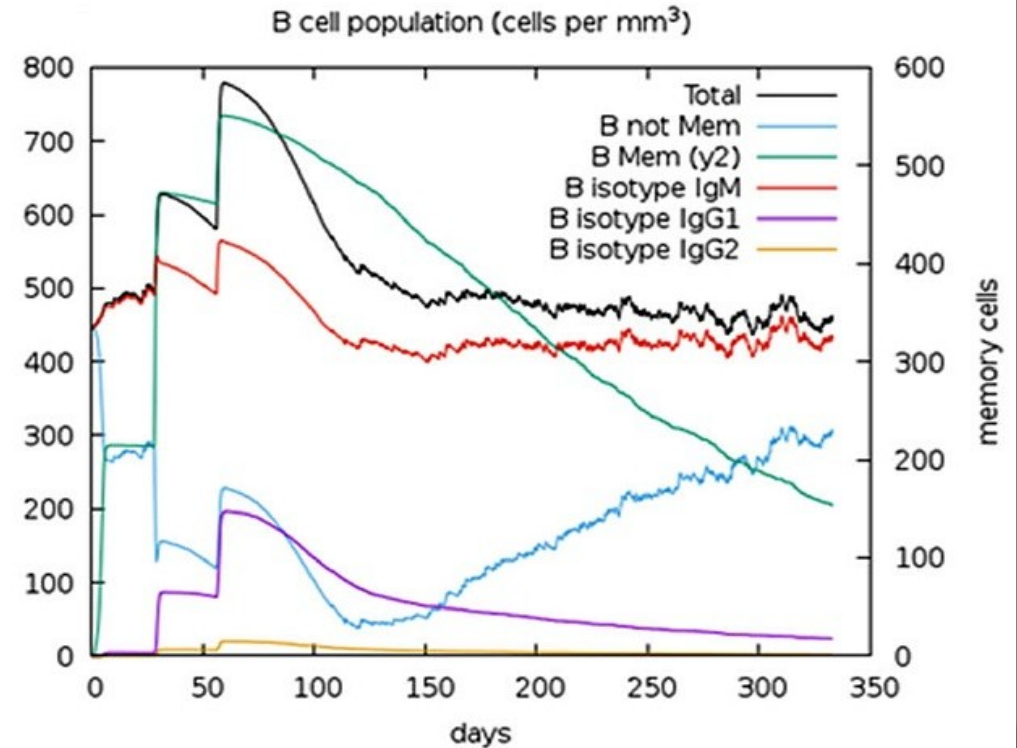
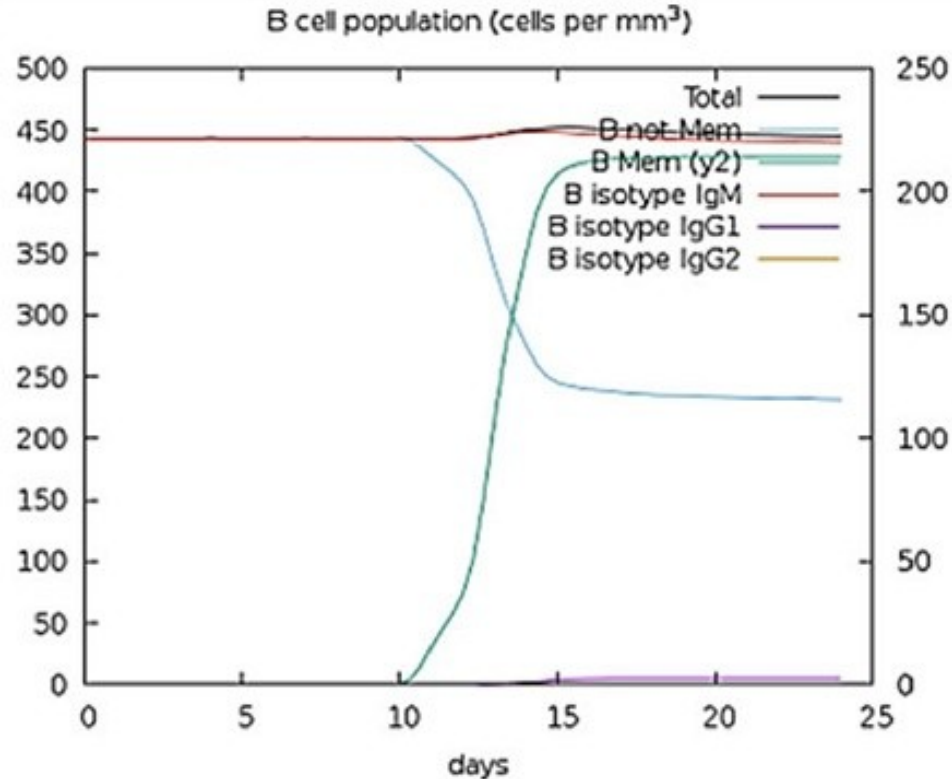
Literature Review: CoV-2 / in-silico

Immune Simulations: Cytokines



Literature Review: CoV-2 / in-silico

Immune Simulations: B Cells



TODO: Image with Vaccine at time = 0

Literature Review:

Recommendations: for PubMed & Journals

- Mark articles with **Caveat-Field:**
 - Docking Multi-Epitope vaccine with MHC-I or II: interesting docking exercise, but functionally completely irrelevant;
 - 12-Dose schedule: NO clinical utility;
 - Conformational Epitopes in Vaccine: **Neo-antigens** => **NO** protection against original virus;

* NO need for retraction: may still be useful exercises to learn Immunology or test in-silico methods / tools;

Pharmacology: CoV-2

- **Pharmacology:**

Majority of PubMed articles with “**in-silico**”

Pharmacology

PubMed Articles:

- 2020: 339 + most of remaining 70;
- Drug-Repurposing & Drug-Libraries;
- Natural-/Phytocompounds;

Molecular Docking:

- Almost ALL articles;

Pharmacology

PubMed Articles:

- 2020: 339 + most of remaining 70;

Summary: based on ~ 50% of articles:

- Phytocompounds: 240; (*)
- Plants: 61; (*)
- Drugs: 140;

Pharmacology: Plants

Plants & Phytocompounds:

- 2020: 339 + most of remaining 70;
- Phytocompounds: 240;
- **Plants:** 61;
- Drugs: 140;

Acacia senegal	Calendula officinalis
Ammoides verticillata	Callophycus oppositifol.
Andrographis paniculata	Camellia sinensis
Annona muricata	Capsicum annum
Artemisia annua	Castanosperm. australe
Asparagus racemosus	Caulerpa racemosa
Astragalus membranac.	Chenopodium quinoa
Brassica napus	Clerodendrum volubile
Broussonetia papyrifera	Crepis sancta

* Summary: based on ~ 50% of articles; final number may be x 2;

Pharmacology: Phytocompounds

Plants & Phytocompounds:

- 2020: 339 + most of remaining 70;
- **Phytocompounds:** 240;
- Plants: 61;
- Drugs: 140;

iridoid	kazinol
isatin	lauruside
isoginkgetin	leupeptin
isoliquiritigenin	limonene
isomesuol	limonoid
jacalin	lividomycin
jamaicin	lupane
kaempferol	macaflavanone
kamalachalcone	maslinic acid

* Summary: based on ~ 50% of articles; final number may be x 2;

Pharmacology: Targets

Targets:

- Main Protease (Mpro/CLpro): > 124;
- PL-Protease: alone or in combination (few);
- Polymerase (RdRp): > 11;
- S or S/ACE2 complex: > 45;
- Other: Methyltransferase, etc.